

10.

Ans: A

The total energy in the oscillation is given by:

$$E_T = \frac{1}{2} m \omega^2 x_0^2 = (0.5)(2)(2\pi(2.5))^2 (0.05)^2 = 0.617 \text{ J}$$

Hence the potential energy is:

$$E_p = 0.617 - 0.36 = 0.26 \text{ J}$$

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Ans: D